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ENERGY295-Electrochemical Energy Storage Systems: Modeling and Estimation

Lecture 10

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Energy Science and Engineering

Electrical Engineering (courtesy)

Precourt Institute of Energy



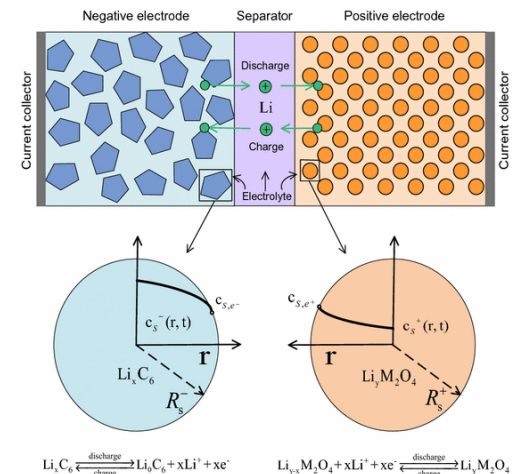
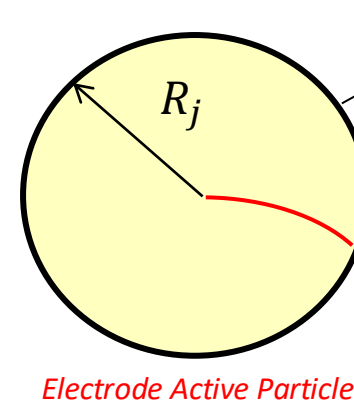
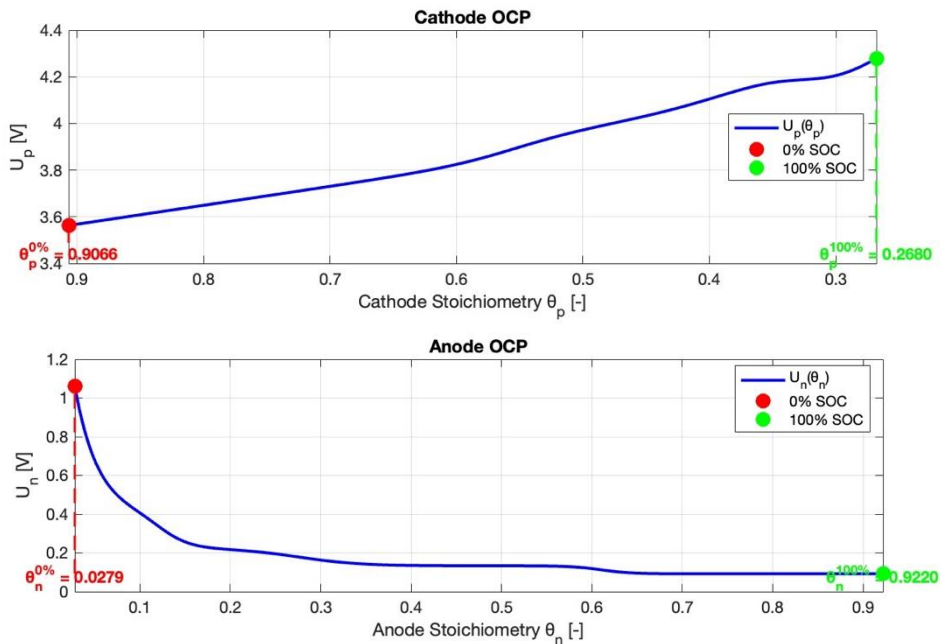
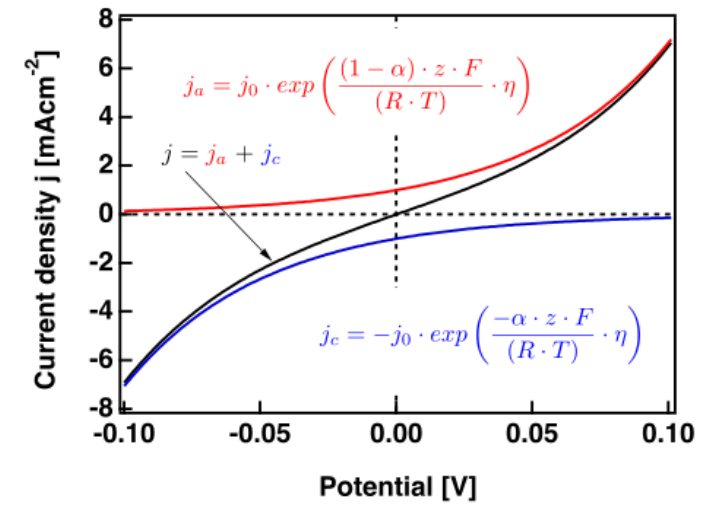
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SPM

The SPM captures battery behavior through

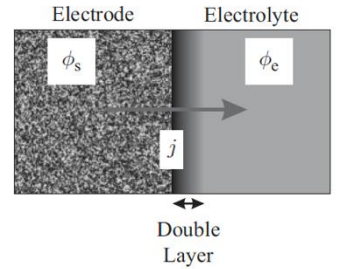
1. thermodynamics (equilibrium voltage),
2. diffusion (lithium transport in the solid phase),
3. reaction kinetics (charge transfer at the interface)



Overpotential

- Overpotential is the extra voltage required to drive electrochemical reactions at a finite rate.
- Overpotential is defined at the electrode–electrolyte interface:

$$\eta_j = \phi_{s,j} - \phi_{e,j} - U_j \quad j \in \{n, p\}$$



$\phi_{s,j}$ = **Solid Phase Potential**: The electrical potential of the solid electrode material (electrons).

$\phi_{e,j}$ = **Electrolyte Phase Potential**: The electrical potential of the liquid electrolyte (ions) at the same point in space.

U_j = **Open Circuit Potential - OCP**: The theoretical equilibrium potential of the active material, which is a function of the state of charge (lithium concentration at the particle surface).

Cell Voltage

Measured terminal voltage:

$$V = OCV + \text{polarization losses}$$

Expanded:

$$V = (U_p - U_n) + (\eta_p - \eta_n) + (\text{ohmic} + \text{transport losses})$$

Kinetic Losses

Exchange Current Density

Exchange Current Density (i_0 (A/cm²)) is the intrinsic rate of an electrochemical reaction at equilibrium, expressed as **current per unit electrode area**

- Even when a battery appears at rest, there is microscopic electrochemical activity at the electrode-electrolyte interface.

Exchange current density measures how fast electrons and ions can transfer across the electrode–electrolyte interface without driving force.

Exchange Current Density

- It tells you how "easy" it is for a specific surface to facilitate a reaction.
 - **High** i_0 : The reaction is kinetically "fast." Even a tiny overpotential will result in a massive flow of current.
 - **Low** i_0 : The reaction is "sluggish". You have to push the system hard with extra voltage just to get it moving.

Butler–Volmer (charge transfer kinetics) at electrode j

- BV describes charge transfer at Electrode solid surface $\leftrightarrow$ Electrolyte interface

For a Lithium-Ion Cell You Have two BV equations

- Negative electrode interface
- Positive electrode interface

$$i_j = i_{0,j} \left[\exp \left(\frac{\alpha_a F \eta_j}{RT} \right) - \exp \left(-\frac{\alpha_c F \eta_j}{RT} \right) \right]$$

Kinetic Parameters

$\alpha_{a,j}$ — Anodic charge transfer coefficient

- Fraction of overpotential driving oxidation
- Units: dimensionless

$\alpha_{c,j}$ — Cathodic charge transfer coefficient

- Fraction of overpotential driving reduction
- Units: dimensionless

Current Term

i_j — **interfacial reaction current density**

- Net electrochemical current at interface
- Units: A/m² (or A/cm²)

$i_{0,j}$ — Exchange current density

- Units: A/m² (or A/cm²)

Driving Force

η_j — Overpotential

- Distance from equilibrium potential
- Units: Volts (V)

F — Faraday constant

- Units: C/mol

R — Gas constant

- Units: J/(mol·K)

T — Temperature

- Units: Kelvin (K)

Butler–Volmer at electrode j

Let's assume **symmetric BV**:

$$\alpha_a = \alpha_c = \alpha_j$$

Substitute:

$$i_j = i_{0,j} \left[\exp\left(\frac{\alpha_j F \eta_j}{RT}\right) - \exp\left(-\frac{\alpha_j F \eta_j}{RT}\right) \right]$$

Then BV becomes:

$$i_j = 2i_{0,j} \left[\sinh\left(\frac{\alpha_j F \eta_j}{RT}\right) \right]$$

Let

$$x = \frac{\alpha_j F \eta_j}{RT}$$

Then:

$$i_j = i_{0,j}(e^x - e^{-x})$$

Use the identity:

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \Rightarrow e^x - e^{-x} = 2\sinh(x)$$

Specific interfacial surface area

$i_j [\text{A/m}^2]$ reaction current density **per active interface area** = How much electrochemical current flows per square meter of electrode–electrolyte interface

- We need to introduce specific interfacial surface area ($[1/\text{m}]$) → How much particle surface area exists inside 1 m^3 of electrode.

$$a_s = \frac{\text{total particle surface area}}{\text{electrode volume}}$$

For spherical particles:

$$a_{s,j} \approx \frac{3\varepsilon_{AM}}{R_j}$$

Volumetric reaction current density

- **Volumetric reaction current density** (usually denoted j_j (is the **electrochemical current generated or consumed per unit electrode volume** due to interfacial electrochemical reactions.
- **How much electrochemical reaction current is being generated per unit volume of electrode** j_j [A/m³]

volumetric reaction
current density
[A m⁻³]

$$j_j = a_{s,j} i_j$$

specific interfacial surface
area (active surface area
per electrode volume)
[m²m⁻³ = m⁻¹]

interfacial reaction current density
(Butler–Volmer current at the
surface) [A m⁻²]

Connect to total cell current I_{batt}

Electrode volume: $V_j = AL_j$

Then:

$$\begin{aligned} I_{batt} &= j_j AL_j \\ &= a_{s,j} i_j AL_j \quad \longrightarrow \quad i_j(t) = \frac{I_{batt}(t)}{A L_j a_{s,j}} \end{aligned}$$

From BV:

$$i_j = 2i_{0,j} \left[\sinh \left(\frac{\alpha_j F \eta_j}{RT} \right) \right] \qquad \sinh \left(\frac{\alpha_j F}{RT} \eta_j(t) \right) = \frac{I_{batt}(t)}{2A L_j a_{s,j} i_{0,j}(t)}$$

Invert $\sinh(\cdot)$

$$\eta_j(t) = \frac{RT}{\alpha_j F} \sinh^{-1} \left(\frac{I_{batt}(t)}{2A L_j a_{s,j} i_{0,j}} \right)$$

Exchange current density

- If $\alpha_j = 0.5$ gives

$$\eta_j = \frac{2RT}{F} \operatorname{asinh} \left(\frac{i_j}{2i_{0,j}} \right)$$

$$a_{s,j} = \frac{3\epsilon}{R_j}$$

$$\eta_j(t) = \frac{2RT}{F} \sinh^{-1} \left(\frac{I_{batt}(t)R_j}{2AL_j3\epsilon i_{0,j}} \right)$$

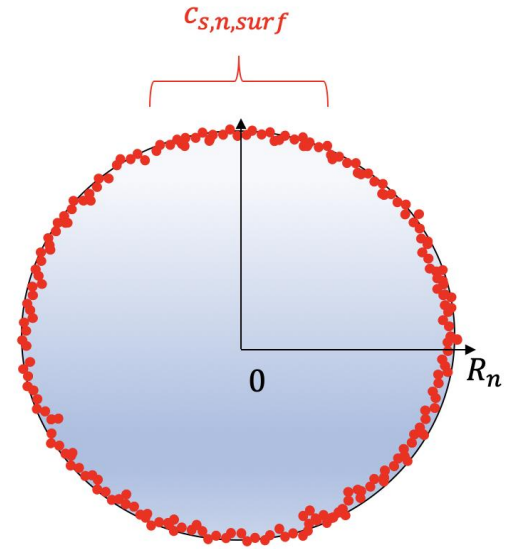
- What about $i_{0,j}$?

Exchange current density

$$i_{0,j} = F k_{0,j} \sqrt{c_e} \sqrt{c_{s,j}^{surf} (c_{s,j}^{max} - c_{s,j}^{surf})}$$

c_e = Average
electrolyte
concentration [mol/m³]

$$\theta_{s,j}^{surf} = \frac{c_{s,j}^{surf}}{c_{s,j}^{max}}$$



Battery Cell Voltage

$$V(t) = U_p \left(\theta_{s,p}^{surf}(t) \right) - U_n \left(\theta_{s,n}^{surf}(t) \right) + \eta_p(t) - \eta_n(t) - I_{batt}(t)R$$

```
%% =====
%% U_p - CATHODE OPEN CIRCUIT POTENTIAL
%% =====
%
% DESCRIPTION:
% Computes the open circuit potential (OCP) of the NMC (Nickel Manganese Cobalt oxide)
% cathode as a function of lithium stoichiometry.
% INPUTS:
% x      - Stoichiometry (dimensionless lithium concentration) [-]
%         Range: 0 (fully lithiated) to 1 (delithiated)
% T_core - Core temperature [K] (not used in current implementation)
% param  - Parameter structure (not used in current implementation)
%
% OUTPUTS:
% y      - Cathode open circuit potential [V] vs. Li/Li+
%
% NOTES:
% - Fitted from experimental data for NMC
```

```
function y = U_p(x, T_core, param)
% Multi-term empirical fit for NMC OCP
% Form: linear + multiple tanh terms

y = -0.8090*x + 4.4875 - ...
    0.0428*tanh(18.5138*(x-0.5542)) - ...
    17.7326*tanh(15.7890*(x-0.3117)) + ...
    17.5842*tanh(15.9308*(x-0.3120));

end
```

```
%% eta_cathode - CATHODE OVERPOTENTIAL (BUTLER-VOLMER)
%% =====
```

```
%
% DESCRIPTION:
% Computes the activation overpotential at the cathode using the Butler-Volmer equation.
% Similar to anode but with opposite sign due to opposite reaction direction.
%
% PHYSICS:
% Same Butler-Volmer form as anode, but:
% - Reaction proceeds in opposite direction
% - Different kinetic parameters (THETA(5), THETA(6))
% - Different geometric parameters (Rsp, Lp, csp_max)
%
% INPUTS:
% x      - Surface stoichiometry [-] (0 to 1)
% T_core - Core temperature [K]
% input_crt - Applied current [A] (positive = discharge)
% param  - Parameter structure with cathode parameters
%
% OUTPUTS:
% eta_p  - Cathode overpotential [V]
%         Positive during discharge (energy gain)
%         Negative during charge
%
```

```
function eta_p = eta_cathode(x, T_core, input_crt, param)
% Thermal voltage: RT/F [V]
V_thermal = param.Rg.*T_core./param.F;

% Butler-Volmer overpotential (note negative sign)
eta_p = 2.0.*V_thermal.*asinh( ...
    -(param.Rsp./(param.k0p.*param.epsp.*param.Lp.*param.Acell.*param.F.*param.csp_max.*sqrt(param.ce))./6.0).*(input_crt./sqrt(x.*(1-x))) ...
);

end
```

SPM Equations (Summary)

- **Solid phase mass transport equation:**

$$\frac{\partial c_{s,j}(r, t)}{\partial t} = \frac{D_{s,j}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_{s,j}(r, t)}{\partial r} \right)$$

$$\text{Boundary Conditions: } D_{s,j} \frac{\partial c_{s,j}(r, t)}{\partial r} \Big|_{r=0} = 0; \quad D_{s,j} \frac{\partial c_{s,j}(r, t)}{\partial r} \Big|_{r=R_j} = \frac{\pm j_j}{a_{s,j} F}$$

- **Cell overpotential:**

$$\eta_j(t) = \frac{RT}{\alpha_j F} \sinh^{-1} \left(\frac{I_{batt}(t)}{2AL_j a_{s,j} i_{0,j}} \right)$$

- **Cell terminal voltage equation:**

$$V_{batt}(t) = U_p(\theta_{surf}^{pos}) + \eta_p(t) - U_n(\theta_{surf}^{neg}) - \eta_n(t) - RI_{batt}(t)$$

- **Bulk state of charge:**

$$SOC_{bulk,n} = \frac{\frac{c_{s,avg,n}}{c_{s,max,n}} - \theta_{0\%}^{neg}}{\theta_{100\%}^{neg} - \theta_{0\%}^{neg}}$$

$$SOC_{bulk,p} = \frac{\frac{c_{s,avg,p}}{c_{s,max,p}} - \theta_{0\%}^{pos}}{\theta_{100\%}^{pos} - \theta_{0\%}^{pos}}$$

State-Space Representation of the SPM Model

- The goal is to represent the ODEs of the SPM model in state-space notation:

$$\dot{x} = Ax + Bu$$

- The discretized ODEs in the anode result in a total of $N - 1$ states, and their dynamics is represented as follows:

$$\dot{x}_1 = \begin{bmatrix} \dot{c}_{s,n,1} \\ \dot{c}_{s,n,2} \\ \dot{c}_{s,n,3} \\ \vdots \\ \dot{c}_{s,n,N-1} \end{bmatrix} = \underbrace{\begin{bmatrix} -2 & 2 & 0 & 0 & \cdots & 0 & 0 \\ 1/2 & -2 & 3/2 & 0 & \cdots & 0 & 0 \\ 0 & 2/3 & -2 & 4/3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -2 \end{bmatrix} \cdot \frac{D_{s,n}}{(\Delta r_n)^2} \cdot \begin{bmatrix} c_{s,n,1} \\ c_{s,n,2} \\ c_{s,n,3} \\ \vdots \\ c_{s,n,N-1} \end{bmatrix}}_{A_{11}} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 2 + \frac{2}{N-1} \end{bmatrix} \cdot \frac{1}{\Delta r_n \cdot F \cdot A_{cell} \cdot a_{s,n} \cdot L_n} \cdot u}_{B_1}$$

- The discretized ODEs in the cathode result in a total of $N - 1$ states, and their dynamics is represented as follows:

$$\dot{x}_2 = \begin{bmatrix} \dot{c}_{s,p,1} \\ \dot{c}_{s,p,2} \\ \dot{c}_{s,p,3} \\ \vdots \\ \dot{c}_{s,p,N-1} \end{bmatrix} = \underbrace{\begin{bmatrix} -2 & 2 & 0 & 0 & \cdots & 0 & 0 \\ 1/2 & -2 & 3/2 & 0 & \cdots & 0 & 0 \\ 0 & 2/3 & -2 & 4/3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & -2 \end{bmatrix} \cdot \frac{D_{s,p}}{(\Delta r_p)^2} \cdot \begin{bmatrix} c_{s,p,1} \\ c_{s,p,2} \\ c_{s,p,3} \\ \vdots \\ c_{s,p,N-1} \end{bmatrix}}_{A_{22}} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 2 + \frac{2}{N-1} \end{bmatrix} \cdot \frac{-1}{\Delta r_p \cdot F \cdot A_{cell} \cdot a_{s,p} \cdot L_p} \cdot u}_{B_2}$$

State-Space Representation of the SPM Model

- The states of the model are:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- The input to the model is:

$$u = I_{app}(t)$$

- The dynamics of the states of the SPM model is represented as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u$$

- The output of the model is:

$$y = V_{cell}(t) = U_{0,p}(x_2) - U_{0,n}(x_1) + \eta_p(x_2) - \eta_n(x_1) - R_c \cdot u$$

- Therefore, the state-space representation of the SPM model is:

$$\begin{aligned} \dot{\mathbf{x}}_1 &= \mathbf{A}_{11} \cdot \mathbf{x}_1 + \mathbf{B}_1 \cdot \mathbf{u} & \mathbf{x}_1 &= [c_{s,n,1} \ c_{s,n,2} \ \cdots \ c_{s,n,N-1}]^T \\ \dot{\mathbf{x}}_2 &= \mathbf{A}_{22} \cdot \mathbf{x}_2 + \mathbf{B}_2 \cdot \mathbf{u} & \mathbf{x}_2 &= [c_{s,p,1} \ c_{s,p,2} \ \cdots \ c_{s,p,N-1}]^T \\ \mathbf{y} &= \mathbf{g}(\mathbf{x}_1, \mathbf{x}_2) - \mathbf{R}_c \cdot \mathbf{u} & u &= I_{app}(t) \\ & & y &= V_{cell}(t) \end{aligned}$$